This can allow you to wall off containers from each other and control which containers can communicate with others.

Secret Management

In this book, we followed many of the twelve-factor app principles.[15] For example, we externalized our app config, making it available as environment variables.[16] However, it turns out that environment variables aren't particularly secure—they're available to the entire process, easily leaked, and violate the principle of least privilege.[17] Docker offers a more secure, built-in option called Docker secrets.[18]

Docker secrets are added to a swarm with the `docker secret create` command (having first targeted a swarm manager). Alternatively, you can specify secrets in your deploy file (in Compose format).[19]

Secrets are encrypted inside Swarm's data structures that store them (encryption at rest), as well as on their entire journey to reach the containers that need them (encryption in transit). They are made available to a container via an in-memory filesystem that's mounted at `/run/secrets/<secret_name>`. Only containers explicitly given access to a secret are able to access it.

There's even a built-in mechanism for rotating secrets,[20] which makes it more likely you'll do the Right Thing and rotate your secrets frequently.

Restarting on Failure

By default, when the process running inside a container terminates, the container is stopped. Sometimes, this behavior is exactly what we want. For example, our `database-migrator` service is supposed to do its job of migrating the database and then exit.

However, what about our Rails app containers running in production? If something goes wrong that causes the app to crash (for example, in the case of a memory leak), it would be nice if the containers themselves could be resilient and handle failures more gracefully. Who wants to be woken in the middle of the night to fix issues?

15. https://12factor.net
16. https://12factor.net/config
17. http://movingfast.io/articles/environment-variables-considered-harmful/
18. https://docs.docker.com/engine/swarm/secrets/
19. https://docs.docker.com/engine/swarm/secrets/#use-secrets-in-compose
20. https://docs.docker.com/engine/swarm/secrets/#example-rotate-a-secret